

MedTrack_DV

(Hospital Operations & Patient Analytics Dashboard)

1. Introduction

The Hospital Operations & Patient Analytics project is a data analytics and visualization solution designed to turn hospital records into an interactive reporting environment. The project focuses on patient activity, admissions and discharges, departmental performance, hospital capacity, staffing, equipment and patient transfers.

The preparation stage was completed using Python and Pandas in Visual Studio Code. The resulting prepared dataset was then used in Microsoft Power BI to build four connected analytical pages: Hospital Overview, Patient Flow, Department Analytics and Resource Utilization.

1.1 Objectives

- Prepare hospital data for consistent analysis.
- Inspect and validate patient, hospital, admission, discharge, department and resource information.
- Engineer operational KPIs that can be monitored interactively.
- Build Power BI dashboards that answer practical hospital-management questions.
- Enable users to move from high-level hospital status to patient-flow, departmental and resource-level analysis.
- Document the complete workflow so the project can be understood and reproduced.

2. Project Workflow

The project follows a sequential analytics pipeline. Raw records are first inspected and prepared, then converted into analytical measures and finally presented through an interactive Power BI report. Each stage provides the input required by the next stage.



2.1 Workflow Explanation

Stage	Explanation
1. Source Data	The project starts with the Hospital Operations Dataset obtained from Kaggle. The data contains hospital, patient, admission, department, capacity, staff, equipment and transfer information.
2. Data Inspection	Python and Pandas are used to load the data and examine its structure, field types, missing values, duplicates and value distributions.
3. Cleaning & Transformation	Fields are standardized and prepared for analytics. Date, numeric and categorical fields are checked so that later calculations and filters behave consistently.
4. KPI Engineering	Operational indicators are derived from the prepared records. These include admission volume, length of stay, occupancy, readmission, bed utilization, staff utilization, equipment availability and transfers.
5. Prepared Dataset	The cleaned and enriched dataset is stored in Excel and becomes the source used for the Power BI report.
6. Power BI Data Model	The prepared dataset is loaded into Power BI and the required fields and measures are made available to the report visuals.
7. Dashboard Development	Four analytical pages are created. Each page addresses a different group of operational questions.
8. Testing & Validation	KPI values, filters, navigation, visual behavior and the interpretation of the dashboard are checked.
9. Final Report	The validated Power BI report provides a single interactive environment for exploring hospital operations.

3. Dataset Overview

The supplied prepared dataset contains 10,000 records and 61 fields. It combines patient-level information with hospital, department, capacity, staff, equipment and transfer attributes.

Dataset characteristic	Value
Rows	10,000
Columns	61
Hospitals	20
Departments	10
States	6
Cities	6
Complete duplicate rows	0

3.1 Main Data Categories

Category	Examples	Purpose
Patient	Patient ID, Gender, Age, Blood Type	Patient-level analysis
Hospital	Hospital Name, Type, City, State	Hospital and geographic comparison
Admission	Admission Date, Admission Type	Admission-volume analysis
Discharge	Discharge Date, Length of Stay	Stay and discharge analysis
Department	Department, Department ID	Department comparison
Beds	Beds Available, ICU Beds, Bed Occupied	Capacity analysis
Staff	Doctor, Nurses, Staff Count	Staffing analysis
Equipment	Equipment ID, Status, Usage Duration	Equipment utilization
Transfers	Transfer From/To, Transfer Date, Transferred	Patient-flow analysis
Derived metrics	Occupancy, readmission, transfer and utilization fields	KPI reporting

4. Python and Visual Studio Code Implementation

Python was used as the scripting layer rather than performing the entire preparation process manually. Visual Studio Code was used as the development environment. Pandas provided the main DataFrame operations for loading, checking and transforming the hospital records.

4.1 Preparation Activities

- Load the Excel dataset into a Pandas DataFrame.
- Inspect the dataset structure and field names.
- Check duplicate records and patient identifiers.
- Review missing values and date fields.
- Validate numerical and categorical data types.
- Create and retain analytical fields required by the dashboard.
- Export the prepared dataset to Excel for Power BI.

The supplied final workbook contains derived fields for admissions, occupancy, bed utilization, readmission, transfers, staff utilization, equipment usage and departmental efficiency. These fields allow the Power BI report to focus on analysis rather than repeating the full preparation process.

5. KPI Engineering

KPI	Meaning	Use in dashboard
Total Admissions	Number of admission records	Overall patient volume
Average Length of Stay	Average hospitalization duration	Patient flow and capacity pressure
Occupancy Rate	Occupied capacity relative to the available capacity basis	Hospital capacity monitoring
Readmission Rate	Readmitted population relative to the	Repeat utilization monitoring

	relevant population	
Bed Utilization Rate	Occupied beds compared with departmental capacity	Bed capacity analysis
Department Efficiency Score	Composite department-performance indicator	Relative department comparison
Staff Utilization Rate	Patient/bed load relative to staff	Staffing pressure
Equipment Availability Rate	Available equipment relative to equipment inventory	Equipment readiness
Transfer Rate	Transferred patients relative to the relevant patient population	Patient movement analysis

5.1 Prepared Dataset KPI Snapshot

Total Admissions: 10000.00

Average Length of Stay: 8.07

Overall Occupancy Rate: 50.36

Readmission Rate: 49.87

Bed Utilization Rate: 50.36

Department Efficiency Score: 12.58

6. Dashboard Suite

The supplied Power BI report contains four report pages. The pages are deliberately separated by business perspective so that the user can start with an overall status and then investigate the specific operational area behind an observation.

Hospital Overview

The Hospital Overview page is the executive entry point. Its purpose is to answer: How is the hospital performing overall? KPI cards provide the immediate status, while trend and comparison visuals help identify changes across time and categories.

Hospital Overview

KPI Cards

Admissions • Occupancy • Readmissions

Trend Charts

Monthly operational movement

Category Comparison

Admission / hospital comparisons

Filters

Year • Location • Hospital • Department

Documentation visual: illustrates the analytical purpose of the report elements.

Visual	Explanation	Question answered
KPI Cards	Condense the main hospital indicators into immediately readable values.	What is the current operational status?
Trend Charts	Show movement across the selected time period.	Is the metric increasing or decreasing?
Category / Hospital Comparisons	Compare admission or performance levels between categories or hospitals.	Where is activity highest or lowest?
Slicers	Restrict the report to a selected context.	What happens for this year, location, hospital or department?

Patient Flow

The Patient Flow page examines what happens to patients during the hospital journey. It focuses on admissions, discharges, length of stay and transfers, allowing users to identify periods or departments associated with higher patient movement.

Patient Flow

KPI Cards

Average Stay • Transfers • Transfer Rate

Admission / Discharge Trends

Patient volume over time

Transfer Analysis

Movement between departments

Stay Analysis

Length-of-stay pattern

Documentation visual: illustrates the analytical purpose of the report elements.

Visual	Explanation	Question answered
KPI Cards	Summarize average stay and transfer indicators.	How long do patients stay and how much movement occurs?
Admission / Discharge Trends	Display patient movement over time.	When are admissions or discharges higher?
Transfer Analysis	Compares patient movement involving departments.	Which areas are involved in more transfers?
Stay Analysis	Shows changes in average length of stay.	How is patient stay duration changing?

Department Analytics

The Department Analytics page compares departments using workload and performance indicators. It helps identify departments with comparatively high or low efficiency, patient volume, occupancy, readmission or transfer activity.

Department Analytics

KPI Cards

Efficiency • Departments • Transfer Rate

Department Comparison

Efficiency / occupancy / workload

Trend Analysis

Patient volume by department

Readmission Analysis

Department-level comparison

Documentation visual: illustrates the analytical purpose of the report elements.

Visual	Explanation	Question answered
KPI Cards	Summarize department-oriented indicators.	What is the current department-level picture?
Department Comparison Charts	Rank or compare departments by performance indicators.	Which departments stand out?
Patient Volume Trend	Shows how department workload changes over time.	When does department activity rise or fall?
Readmission Analysis	Compares repeat-use behavior across departments.	Where is readmission activity relatively high?

Resource Utilization

The Resource Utilization page examines how hospital resources are being used. Beds, staff and equipment are presented as operational capacity measures so that resource pressure and availability can be compared.

Resource Utilization

KPI Cards

Bed • Equipment • Staff utilization

Bed Analysis

Capacity and occupied beds

Equipment Analysis

Status and usage

Staff Analysis

Staff utilization by department

Documentation visual: illustrates the analytical purpose of the report elements.

Visual	Explanation	Question answered
KPI Cards	Summarize bed, staff and equipment measures.	What is the current resource position?
Bed Analysis	Compares available and occupied capacity.	Where is bed pressure highest?
Equipment Analysis	Shows equipment status or usage patterns.	Which resources are available or heavily used?
Staff Analysis	Compares staff utilization across departments.	Where is staff load relatively high?

7. Dashboard Interaction and Navigation

- Use the slicers to define the analysis context before interpreting KPI cards.
- Hover over charts to inspect individual categories or time periods.
- Use the report navigation to move between hospital-wide, patient-flow, departmental and resource perspectives.
- When an unusual KPI appears, move to the dashboard that contains the dimensions needed to explain it.
- Clear or reset filters when a full-dataset view is required.

The intended analytical path is: Overview → identify a pattern → select the relevant detailed page → compare dimensions → investigate the operational driver. This is more useful than treating each visual as an isolated chart.

8. Testing and Validation

Test area	Validation objective
Data validation	Confirm that the prepared dataset loads correctly and contains the required fields.
KPI validation	Check that dashboard measures are consistent with the prepared data.
Filter validation	Confirm that slicers change the intended visuals.
Navigation validation	Confirm that report pages can be accessed through the navigation controls.

Visual validation	Check that charts, cards and labels render correctly.
Usability validation	Confirm that the dashboard supports the intended analytical workflow.

9. Limitations and Interpretation Notes

- The dataset is an analytical project dataset and should not be treated as a live clinical information system.
- Missing dates can affect time-series and stay-duration calculations for the affected records.
- Percentage KPIs should always be interpreted with their filter context and denominator.
- The dashboard identifies patterns and comparisons; it does not establish causal relationships.
- The report should support operational investigation rather than replace clinical or administrative decision processes.

10. Conclusion

The completed solution demonstrates an end-to-end hospital analytics workflow: source data is prepared with Python, transformed into analytical indicators, loaded into Power BI and presented through four interactive dashboard perspectives. The workflow diagram documents how each stage connects to the next, while the dashboard explanations show how individual visual types contribute to operational analysis.

The strongest feature of the project is the ability to move from a summary KPI to a more specific operational view. Hospital Overview provides the starting point; Patient Flow explains patient movement; Department Analytics highlights differences between departments; and Resource Utilization focuses on capacity, staff and equipment.

11. Tools and Technologies

Tool	Purpose
Python	Data preparation and scripting
Pandas	DataFrame processing and validation
Visual Studio Code	Python development environment
Microsoft Excel	Prepared dataset storage
Microsoft Power BI	Data modeling, KPI reporting and dashboard visualization
Kaggle	Source dataset

12. References

Kaggle Hospital Operations Dataset:

Dataset source:- <https://www.kaggle.com/datasets/amitbakde/hospital-operations-dataset>

GitHub link :- <https://github.com/springboardmentor09876x-cmd/Hospital-Operations-Patient-Analytics-/tree/vineela>